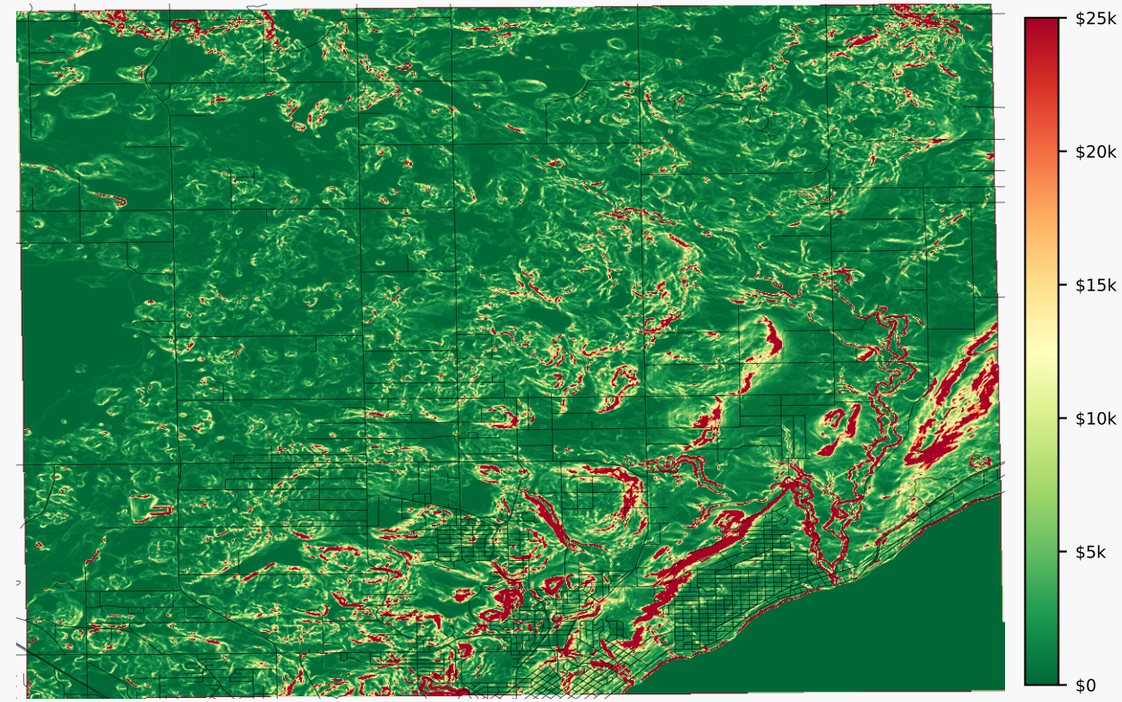
Lab 9 | Cost Surfaces

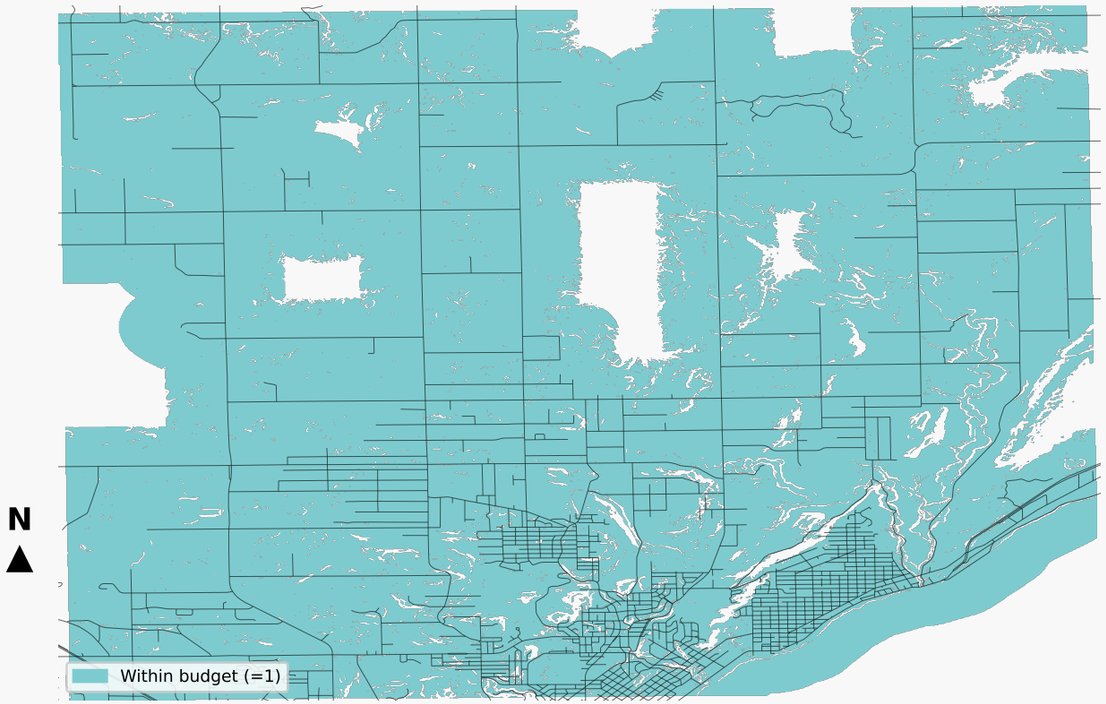
Road Building Costs in Duluth, MN considering slope and distance (within a budget)

Distance Cost

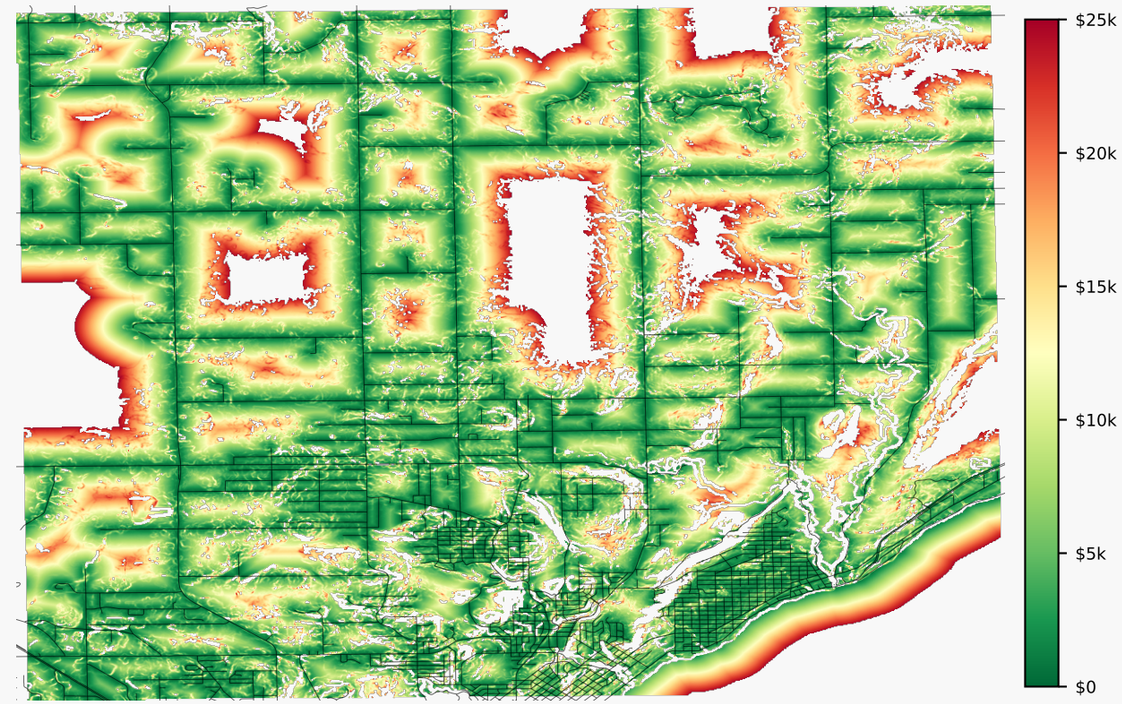
Slope Cost



Budget Mask (≤ $25,000)



Final Cost (within budget)



DistCost = dist × $25/m | SlopeCost = (exp(slope%×0.2)×1000)−1000
TotCost = DistCost + SlopeCost | Mask: TotCost ≤ $25,000 → 1, else NoData
FinalCost = Mask × TotCost | EPSG:26915, 15 m resolution (downsampled from 3 m)